Exercise - Week 13: MATH1061/7861

Duration: 30 minutes

2 Questions - 10 marks

Topics: Inclusion/Exclusion; Introduction to Graph Theory

Student Name:

Student number: |__|__|__|__|__|__|__|__|

*Your student number should be 8 digits and **not** start with an s. e.g. 41234567*

(1) **(5 marks)** A system generates user IDs from the integers 1 to 1000. Due to legacy constraints, certain IDs are flagged as follows:

- IDs divisible by 2 are flagged as *internal*,
- IDs divisible by 3 are flagged as *test*,
- IDs divisible by 5 are flagged as *archived*.

A developer defines a *clean ID* to be an ID that is not flagged in any way.
How many clean IDs are there between 1 and 1000?

(2) **(5 marks)** A graph has 10 edges and 6 vertices. Five of the vertices have degree 3.

(a) What is the degree of the remaining vertex? Explain your reasoning and draw a picture of such a graph.

(b) Does the graph you drew have a circuit?

(c) Does the graph you drew have a Eulerian trail?